PC SALES DATA WAREHOUSE RESEARCH

THE IMPACT OF LEFT JOIN VS INNER JOIN ON FACT TABLE ROW COUNT, DATA ACCURACY, AND STAR SCHEMA INTEGRITY

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RESEARCH FOCUS

This research evaluates how LEFT JOIN and INNER JOIN influence fact table row count, data completeness, data accuracy, and overall star schema effectiveness within a PC Sales Data Warehouse project.

Introduction

In data warehouse design, the structure and accuracy of the fact table are critical to ensuring reliable analytical outcomes. One important decision during the ETL process is the choice of join strategy used when loading transactional data from the staging layer into the star schema, as this directly affects row count integrity, dimensional accuracy, and overall warehouse performance. This research investigates the impact of LEFT JOIN and INNER JOIN within the PC Sales Data Warehouse project, using practical implementation results to evaluate how each approach influences data completeness, schema integrity, and the effectiveness of the final analytical model.

This research highlights:

- Purpose of star schema
- Purpose of fact table
- Data accuracy
- Row count behaviour

Purpose of Star Schema Design

A properly designed star schema improves data accuracy and integrity, reduces redundancy, and enhances system performance.

Purpose of fact table

The fact table stores quantitative, numerical data (measures) and foreign keys to dimension tables. While the fact table grain defines the precise level of detail for a single row in a fact table.

- Our original dataset has 10 000 rows where it references one row per transaction; therefore, the fact table will be expected to have 10 000 rows
- Row count matters because if the fact rows exceed source transactions, it means that data duplication exists, and if the rows are too low, then data loss exists

INNER JOIN

Returns only rows where matching records exist in both staging and dimension tables.

Advantages

- Only valid foreign keys enter the fact table
- No null dimension references
- Protects analytical trust
- Prevents duplication

Potential Risks

- If a dimension match fails, the rows will be excluded in the fact table; this may reduce the row count below the source count

LEFT JOIN

Returns all staging rows, even if dimension matches fail

Advantage

- Preserve all source records

- Missing dimension values can be replaced with an “Unknown” key using COALESCE (COALESCE(customer_id, -1) AS customer_id) (Kimball & Ross, 2013 (Chapter 1-2))

Risk

- If the dimensions contain duplicate matches, 1 staging row = multiple dimension matches
- Unmatched rows remain in the fact table with unknown values
- Too many unknowns may reduce reporting quality

Analysis

Row count using INNER JOIN

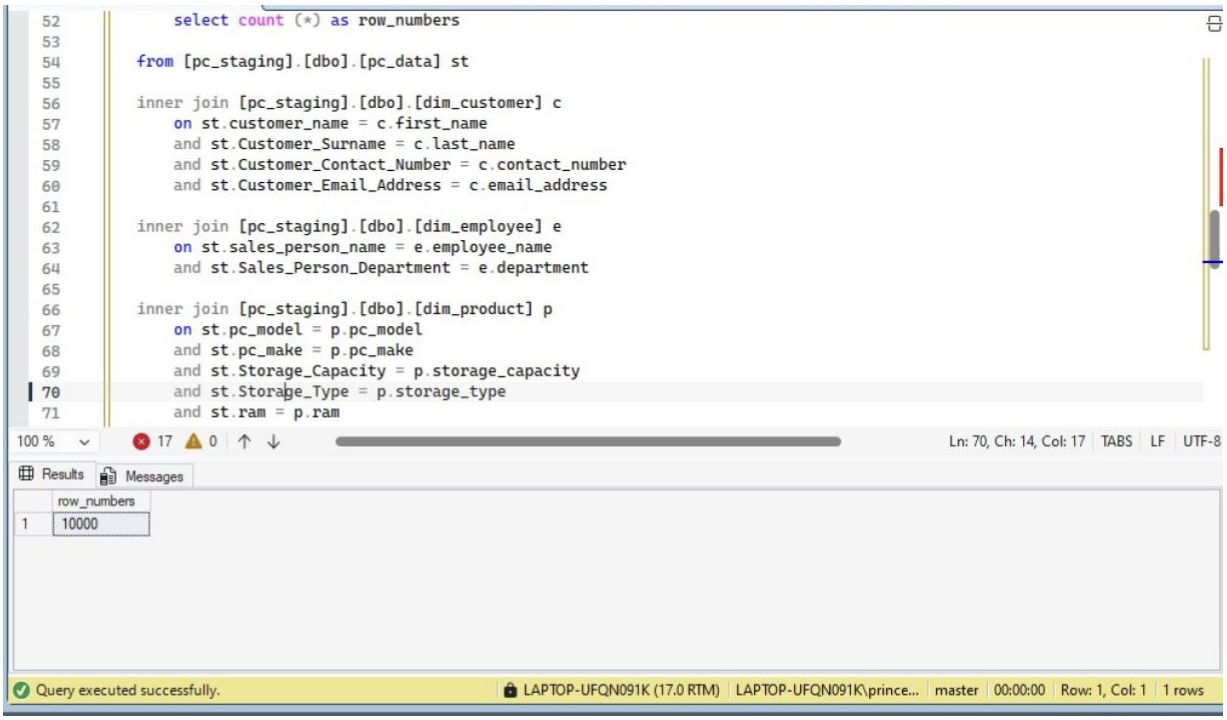

Figure 1: How row count gets affected by INNER JOIN (10 000 rows) (T. Tshikosi).

Row count using LEFT JOIN

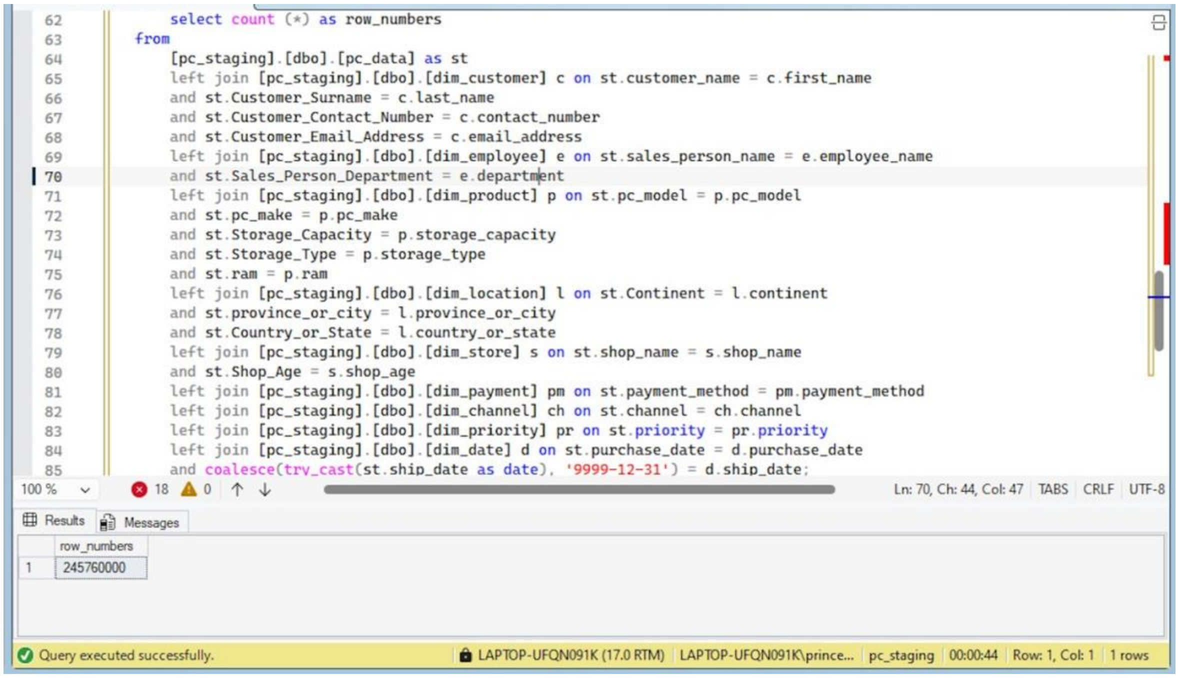

Figure 2: How row count gets affected by LEFT JOIN (245 760 000 rows) (T. Tshikosi).

When using LEFT JOIN row multiplication occurred because dimension tables did not enforce a fully unique business key match for every staging record.

JOIN Strategy Selection

The source dataset is structured as one row per transaction, where each row represents a single PC sales event, including new sales, repairs, returns, or second-hand sales. There is no order-level structure, aggregation, or duplicate records in the staging layer, meaning the fact table should load at a strict 1-to-1 ratio from source to warehouse, with each staging row producing exactly one fact row (Kimball & Ross, 2013). Since the dataset already exists at the correct transactional grain, the fact table's role is to preserve each business event accurately while enforcing valid dimensional relationships.

In this project, INNER JOIN was the most appropriate strategy because it ensured that only records with valid dimension matches were loaded, thereby preserving data integrity and supporting accurate analytical outcomes. While LEFT JOIN was useful during validation for identifying unmatched records, its use in the final fact table load introduced row multiplication due to duplicate dimensional matches, reducing row count accuracy and analytical reliability. Therefore, INNER JOIN provided the more suitable final implementation by preserving transactional accuracy, maintaining schema integrity, and supporting reliable warehouse performance.

Conclusion

An INNER JOIN was the most appropriate final implementation strategy for the PC Sales Data Warehouse because it preserved transactional grain, maintained row count integrity, and ensured valid dimensional relationships. While LEFT JOIN supported validation and detected incomplete records, its susceptibility to row multiplication under non-unique dimension conditions made it unsuitable for final fact table loading.

References

Kimball, R. & Ross, M. (2013). *The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling.* 3rd ed. Wiley. Chapter 3 covers transactional fact tables specifically.